

IT1244: Astral Object Clustering

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Introduction

With the massive volume of data being observed in modern large-scale sky surveys, manual classification of astral objects is becoming increasingly laborious.

In this work, we aim to explore the use of unsupervised learning methods to cluster and classify astral objects based on their photometric and spatial features, and evaluate the effectiveness. This enables scalable analysis of large astral datasets, enabling efficient clustering of astral objects.

Prior studies have explored unsupervised methods for clustering and classification of astral objects. Logan & Fotopoulou (2020) combined Principle Component Analysis (PCA) with Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) on photometric data and demonstrated that density-based clustering can capture non-linear structure on astronomical feature spaces. Their results also highlighted sensitivity to hyperparameter selection and degradation under overlapping class distribution, which are common in photometric datasets. Another study explored using Gaussian Mixture Models (GMM) in X-ray astronomy classification (Pérez-Díaz et al., 2024), showing that probabilistic clustering can model soft class boundaries and uncertainty in assignments.

Dataset

Each observation in the dataset consists of spatial coordinates (ra, dec), photometric magnitudes (u, g, r, i, z) and redshift. Class values are only used for evaluation of various algorithms. During data preprocessing, rows containing invalid placeholder values (-9999) were removed to prevent distortion in distance-based clustering algorithms.

Exploratory data analysis revealed strong correlations among photometric features through a heatmap (Figure A1), indicating redundancy and reinforcing the need to use color indices, such as u-g and g-r. The u-g distribution shows partial separation between classes, with QSOs appearing bluer and galaxies redder, although significant overlap exists (Figure A2). Similarly, the colour-colour

plot (u-g vs g-r) reveals clustering tendencies of the 3 classes but still shows substantial overlap, indicating that colour features alone are insufficient (Figure A3). In contrast, incorporating redshift significantly improves separability, with stars concentrated at zero and QSOs spread over higher values (Figure A4). To improve visualisation clarity, stratified sampling (maximum 1500 samples per class) was applied for plotting, preventing dominant classes, like galaxies, from obscuring patterns.

We applied dimensionality reduction with PCA, which showed that 6 components retain 90% of the dataset's variance (Figure A5). These 6 principal components will be used in subsequent modelling. Overall, the dataset presents challenges such as feature redundancy and overlapping distributions, driving the need for clustering methods with fewer geometric assumptions about cluster structure.

Methods

K-Means Clustering

K-Means clustering assumes clusters are spherical, similarly sized and well-separated in feature space. It partitions data by iteratively assigning each point to its nearest centroid and updating cluster centres to minimize within-cluster variance. In this task, K-means will identify 3 clusters. The choice of $k=3$ is supported by a small but noticeable bend in the elbow plot of inertia (Figure B1), as well as the knowledge that there are 3 astral object classes. The model was run with 20 random initialisation to mitigate convergence to suboptimal local minima.

HDBSCAN

HDBSCAN focuses on areas with high density, allowing it to identify clusters of any shape. Data points in regions of low density will be seen as 'noise', allowing us to effectively filter out any outliers. We cluster using the best parameters for 'min_cluster_size' and 'min_samples' identified through grid search.

Gaussian Mixture Model (GMM)

GMM is a probabilistic model used for clustering that assumes an underlying gaussian distribution of the data. It allows for soft assignments, making it more flexible and does not confine the clusters into fixed shaped clusters.

Results

The presence of true class labels motivates the use of Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) to compute similarity scores for in the evaluation of our clustering algorithms. ARI will be used as the primary evaluation, with NMI as a secondary reference as ARI adjusts for chance agreement, providing a more reliable evaluation in the presence of class imbalance, with galaxy making up 84.5% of the dataset.

Method	ARI	NMI
K-Means	0.373 (3s.f.)	0.283 (3s.f.)
HDBSCAN	0.515	
N	(3s.f.)	0.341 (3s.f.)
GMM	0.411 (3s.f.)	0.351 (3s.f.)

Table I: Comparison of Results

Overall, HDBSCAN achieves the highest ARI (0.515), indicating the strongest alignment with the ground-truth class labels. GMM produces the highest NMI (0.351), while K-Means yields the lowest performance across both metrics.

K-Means yields the poorest results with the lowest ARI but the positive score indicates that it can recover a meaningful part of the class structure despite class imbalance and overlapping features. Based on dominant counts from the contingency table (Figure B2), we can deduce that cluster 2=GALAXY, cluster 0=QSO, and thus cluster 1=STAR. The resulting row-normalised confusion matrix (Figure B3) shows that K-Means successfully distinguished galaxies (88%) and QSOs (86%). However, K-Means failed to separate stars from galaxies, possibly due to their overlapping photometric properties.

The row-normalised contingency matrix (Figure B6) shows that HDBSCAN successfully isolated a pure QSO (cluster 0) and a large, high-purity galaxy (cluster 1). However, stars were not separated from galaxies, indicating that our PCA dataset lacks the resolution to differentiate between the 2 classes. A large portion of the

data points were labelled as noise (cluster -1) by the model, reflecting HDBSCAN prioritized cluster purity over data coverage.

GMM correctly isolates most of the STAR and QSO populations (Figure B7), indicating that the probabilistic mixture model is able to capture a substantial portion of the underlying class structure. However, it exhibits notable contamination within the predicted STAR cluster, where 1,536 GALAXY instances are misclassified, which could be due to the residual overlap in the feature space after PCA.

HDBSCAN yields the strongest results in terms of ARI, since it is a density-based algorithm that can handle outliers and irregular clusters without forcing all points into a group. In contrast, K-means assumes spherical cluster shape while GMM assumes Gaussian distributions. Thus, flexibility in cluster assumptions directly correlates with performance.

Discussion

The unsupervised techniques evaluated demonstrate promising performance for large-scale astral datasets. Although such methods are not typically associated with direct social or ethical concerns, they should be treated as supportive tools rather than definitive sources of truth. The results indicate that models are imperfect and prone to misclassification, due to limitations in data quality and model fitting.

Beyond technical performance, interpretability has implications for how such models are integrated into scientific workflows and how their outputs are trusted. Methods with more explicit geometric or density-based formulations, such as K-means and HDBSCAN, can be more readily inspected and understood, whereas complex probabilistic models may obscure the decision process. This difference affects not only scientific transparency but also the ease with which results can be validated and communicated within research teams. In conclusion, these methods are augmentation tools rather than replacements for domain experts. They can reduce the manual burden of large-scale catalogue inspection and preliminary classification, but expert oversight remains essential for validation.

References

Hu, C.; D’Isanto, A.; Williams, M.; and Polsterer, K. L. 2024. Self-supervised Representation Learning for Astronomical Images with Spectroscopic Labels. Monthly Notices of the Royal Astronomical Society 528(3): 4852–4864.

Logan, C. H. A.; and Fotopoulou, S. 2020. Unsupervised Star, Galaxy, QSO Classification - Application of HDBSCAN. Astronomy & Astrophysics 633(A154).

Appendix A

Exploratory Data Analysis Visualisations

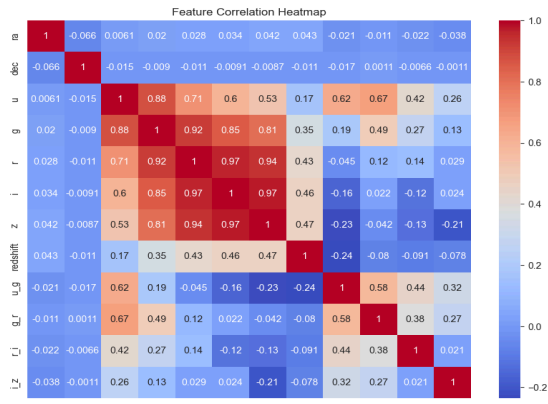


Figure A1: Correlation Heatmap of Features

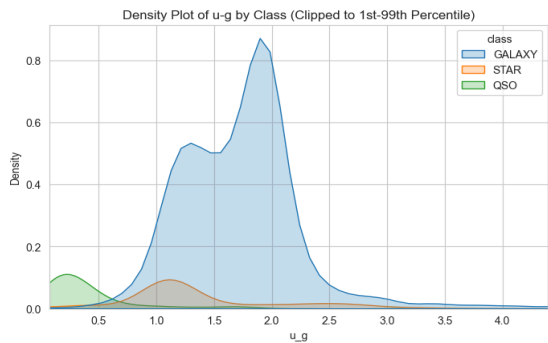


Figure A2: Density Plot of u-g by Class

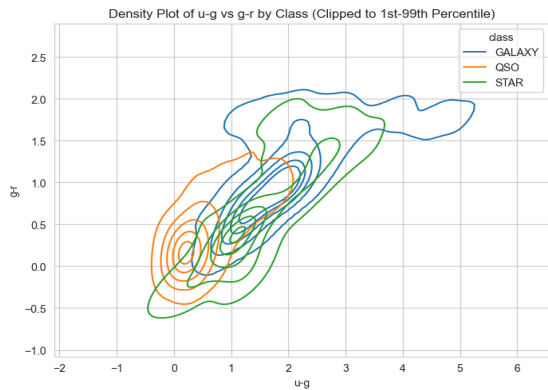


Figure A3: 2D Density Plot of u-g vs g-r by Class

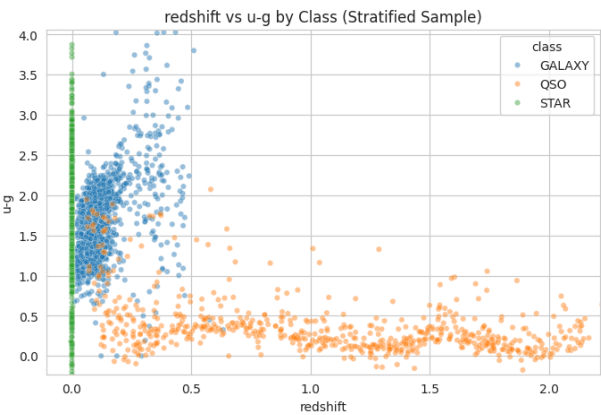


Figure A4: Scatterplot of Redshift against u-g by Class

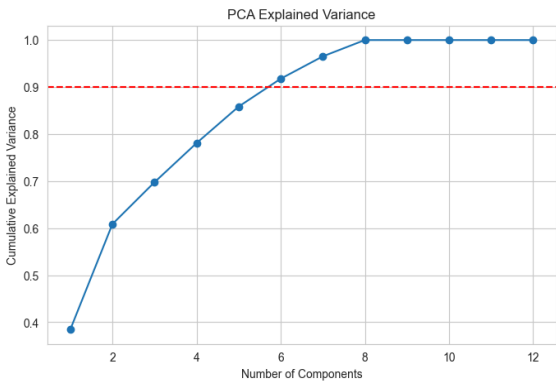


Figure A5: Cumulative Explained Variance from PCA

Appendix B

Unsupervised Clustering Evaluation Plots

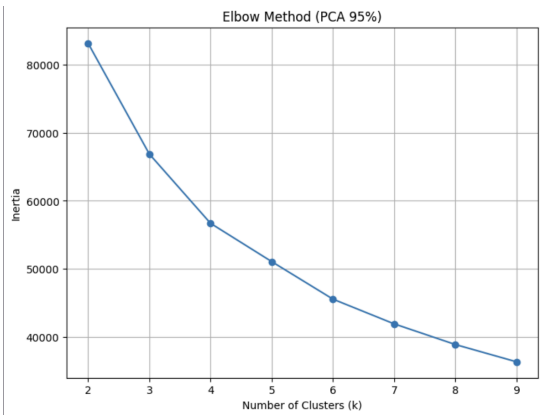


Figure B1: Elbow plot of inertia for k=2 to k=9

True Class	Cluster	Cluster	Cluster
	0	1	2

GALAXY	150	1035	7250
QSO	605	13	67
STAR	200	150	515

Figure B2: Contingency table of true astronomical classes vs K-Means cluster assignments

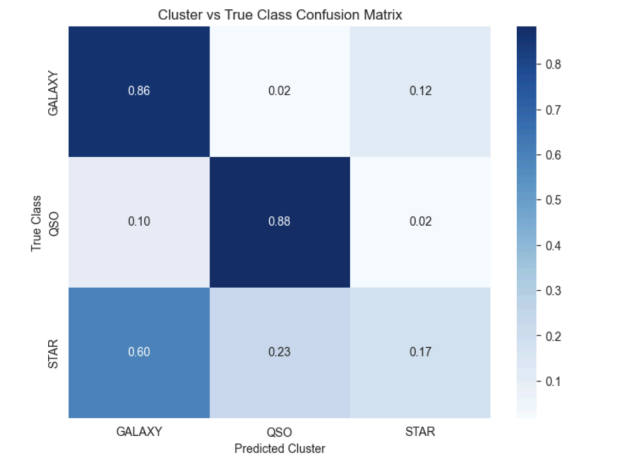


Figure B3: Row-normalised confusion matrix for K-means (proportions per true class)

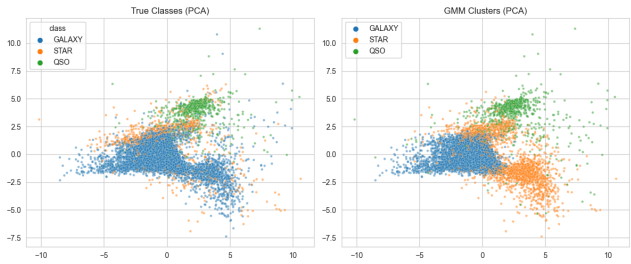


Figure B5: PCA projection of the dataset showing true class labels (left) and GMM clustering results (right)

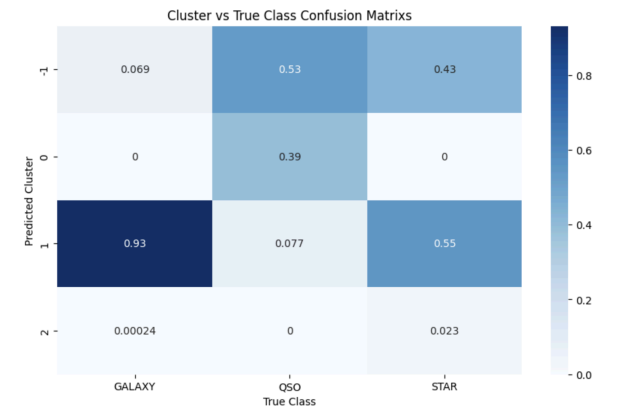


Figure B6: HDBSCAN Row-normalised confusion matrix (proportions per true class)

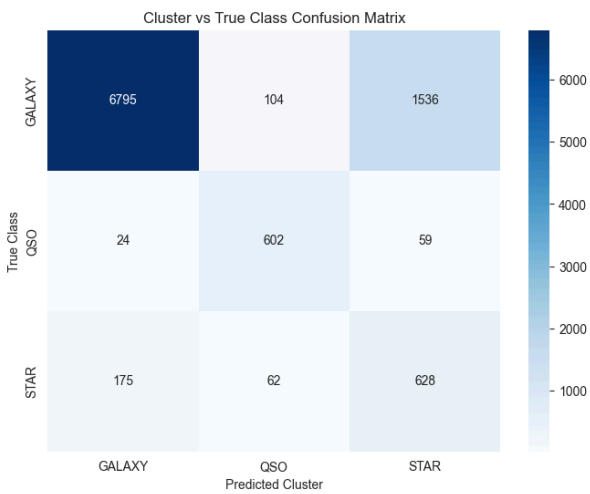


Figure B7: GMM confusion matrix (raw counts)

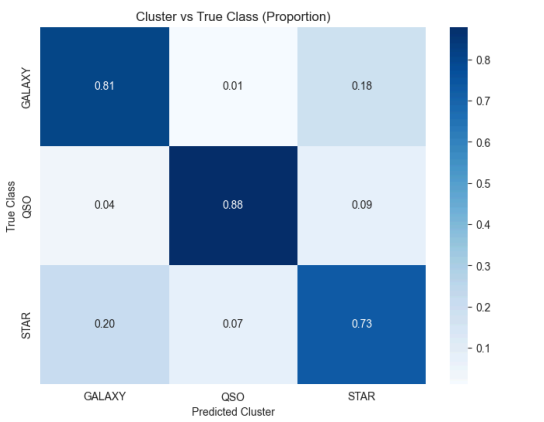


Figure B8: GMM Row-normalised confusion matrix (proportions per true class)

AI Usage Declaration

AI tools (e.g., Claude, ChatGPT) were used to support this project primarily in brainstorming ideas, debugging code, and improving the clarity and structure of written explanations. Core implementation, analysis, and interpretation of results were carried out independently by the team.